

Rishindra Davuluri

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EDUCATION

The University of Texas at Austin	M.S., Computer Science	May 2028
Texas CSB Honors Program	B.S. Computer Science Honors, B.B.A. Business Honors	May 2027

Relevant Coursework: Machine Learning II (H), Principles of ML I, Algorithms (H), Java Data Structures (H), C++ Operating Systems (H), Computer Architecture w/ C (H), Object-Oriented Programming, SQL MIS (H), Data Science in R (H), Decision Science (H), Discrete Math (H), Linear Algebra, Multi-variable Calculus, Statistics & Probability **(H): Honors**

EXPERIENCE

Loopfour (YC W23) - AI Engineer Intern; Mountain View, CA Jun 2026 - Present

Moody College of Communication - Undergraduate NLP Researcher; Austin, TX Nov 2025 - Present

- Categorized narratives via anchor-based semantic classification using **sentence embeddings** and **cosine similarity**
- Engineered **BERTopic** and **RegEx** feature-engineering pipeline to analyze demographic disparities in financial complaints
- Built and deployed an **OpenAI** chatbot survey platform (**React**, **FastAPI**, **AWS**) for a research study, storing responses in **S3**

UT Austin: Learning Directed Operating System - Undergraduate ML Researcher; Austin, TX Oct 2025 - Feb 2026

- Designing low-latency **Model Context Protocol** in **C** to synchronize decision-making across learned OS components
- Benchmarking protocol on **CloudLab** bare-metal instances using conflicting learned agents (e.g., Prefetcher vs. Allocator)

Capital One - Software Engineer Intern; McLean, VA Jun 2025 - Aug 2025

- Architected cache library and **Snowflake** dataset using **Python** and **AWS Lambda** for omni-channel experience
- Integrated cache into **Vue.js** and **Vuex** webpage, improving workflow completion by **40%** for **80K** monthly fraud concerns
- Built educational phishing simulation platform using **Next.js** and **Twilio** to send custom emails and track user clicks

UT Austin: Dynamic Belief Games - Undergraduate ML Researcher; Austin, TX Sep 2024 - Jan 2025

- Trained and tested **GCNNs** and **PyTorch SB3** reinforcement learning models to optimize network communication
- Leveraged **NumPy**, **Pandas**, and **Matplotlib** to present data analysis to **U.S. Army Futures Command's UTDD**

Fidelity Investments - Software Engineer Intern; Westlake, TX Jun 2024 - Aug 2024

- Enhanced batch responsible for **100M+** updates to central DB with **AWS Q Developer**, **SQL**, & **MyBatis**
- Accelerated processing by **20%** by migrating 30+ files from **C++** to **Java** for multi-threading and cloud integration

PROJECTS

Ace: Proactive AI Study Agent | Go, Python, Gemini API, Angular, gRPC, Google Cloud, Neo4j, BigQuery

- Architected AI tutoring microservice (**Go**, **Python gRPC**, **Angular**) deployed on GCP with **Neo4j** and **BigQuery** for telemetry
- Engineered **Gemini RAG** pipeline with **LLM-as-a-judge** guardrails, quiz generation, and Google Calendar OAuth scheduling

Operating System & Graphics | C++, Python, Assembly, QEMU, Bash, Linux, Multi-threading

- Developed core **C++** and **Assembly** kernel: interrupt handling with $O(1)$ helper threads; fork/wait system calls; semaphores
- Engineered virtual memory, **Ext2** file system with third-level indirection, and VBE graphics subsystem with **Python** frame conversion

RestoreAI | Python, Pandas, NumPy, Random Forest, Gemini API, React, JavaScript, RESTful APIs

- Selected **Random Forest** model in Python using **Pandas** and **NumPy** after cross-validation and hyperparameter tuning
- Streamlined React UI to ML model integration using **RESTful API** architecture and **Gemini API** function calling

LEADERSHIP EXPERIENCE AND ACTIVITIES

Texas Product Engineering Organization - Engineering and Product Fellow; Austin, TX Sep 2024 - Present

- Developing **Python** web crawlers, **PHP** plugins, and websites with **React**, **REST**, and **Vercel** for nonprofits
- Built **Locus** with **TS**, **Node.js**, and **Firebase** (**Realtime DB/Firestore**) for **sub-100ms** location and status updates

Texas Infinites - Gender Inclusive Spirit & Volunteer Organization, Recruitment Committee; Austin, TX Jan 2025 - Present

ADDITIONAL INFORMATION

Technical Languages: Python, C++, TypeScript, Go, Java, C, JavaScript, C#, R, SQL, SOQL, CQL

Cloud & DB: GCP (GCS, GC Run, BigQuery, Firebase), AWS (Lambda, S3, EC2), Terraform, Snowflake, Neo4j, Apache Cassandra

Tools: PyTorch, NumPy, Pandas, sklearn, Matplotlib, Angular, React, FastAPI, Salesforce, Linux, Docker, GitHub Actions

Certifications: Eagle Scout | Azure AI Fundamentals | NLP & Text Mining (NLTK, BERTopic) by Simplilearn

Service & Interests: Singing & Guitar; Camping; Portrait Drawing; Volleyball; Soccer

Languages: Native in English and Telugu; Conversant in Spanish